Week 3

Axaitech

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01 Challenges

Challenges

Struggles with features

Had very few features; unable to pull more information from the data (at the time!)

Low modelization accuracy

Modelization that we used had an accuracy of around 73%--too little features paired with too little data

Unable to build peptide with model

Due to issues with new dataset, we were unable to build a peptide; additionally, our modelization accuracy was low, so our peptide would not have been successful

02 Feature Extraction

Feature Extraction

Originally

Had 14 columns, ended with around 1.9K

Sequence analysis

Used Biopython to add different columns of sequence analysis

PSSM

Scored particular amino acids at specific positions

03 Modelization

Random Forest Model

```
Accuracy: 0.9375
Precision: 0.9683
Recall: 0.9242
F1-Score: 0.9457

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.96      0.93        46
           1       0.97      0.92      0.95        66

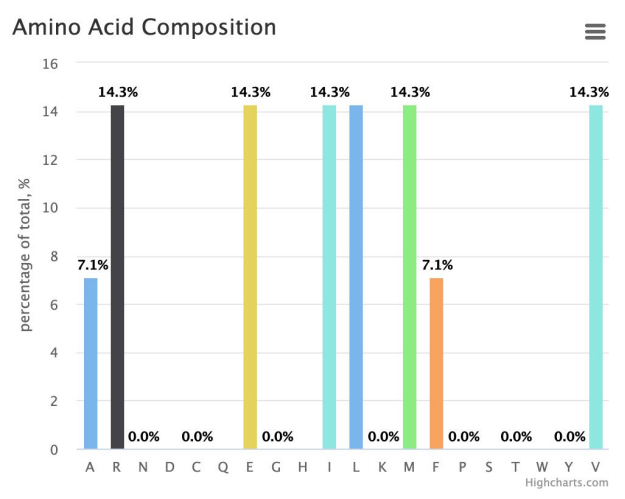
    accuracy                           0.94       112
   macro avg       0.93      0.94      0.94       112
weighted avg       0.94      0.94      0.94       112
```

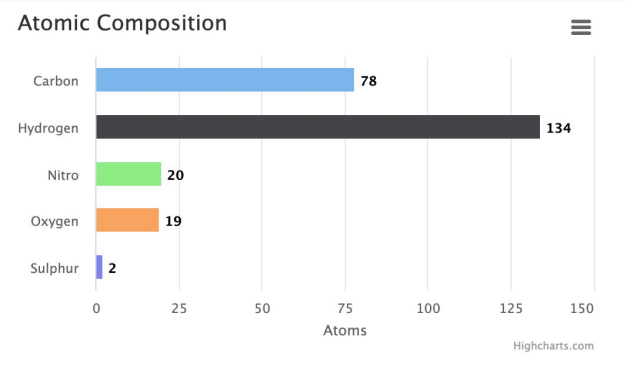
- Accuracy of the model has gone from 73% to 94%
- Features include, Molecular Weight, GRAVY, pI, boman, Aliphatic, Helix Fraction, Sheet Fraction, Coil Fraction and Sequence features
- Sequence features break down each sequence into percentage of amino acids and position in the sequence (position-specific scoring matrix)

04 New Peptides

New Peptide #1

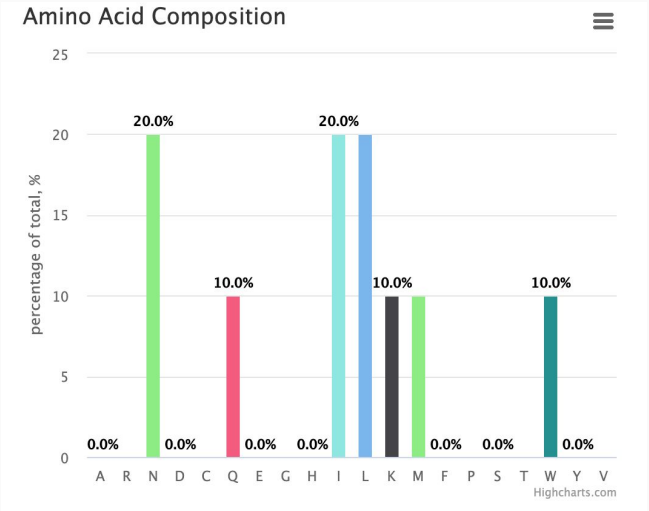
Sequence: VAVERIRILMFELM
Length: 14
Molecular Weight: 1720.17
pI: 7.15
Charge: 0.002
GRAVY: 1.2429
Aliphatic Index: 160
Boman Index: 0.4436

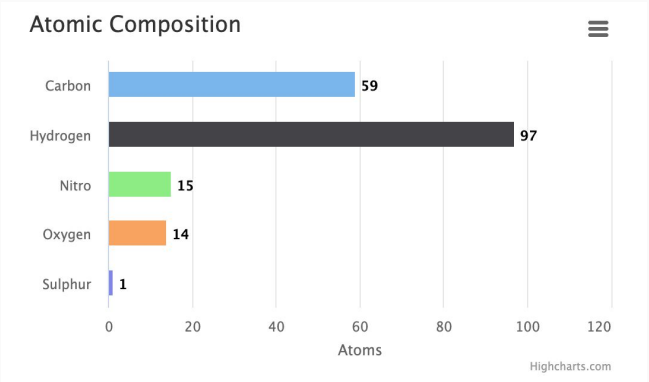




New Peptide #2

Sequence: NNKLMLIWQI
Length: 10
Molecular Weight: 1272.57
pI: 9.67
Charge: 0.984
GRAVY: 1.32
Aliphatic Index: 156
Boman Index: 0.001

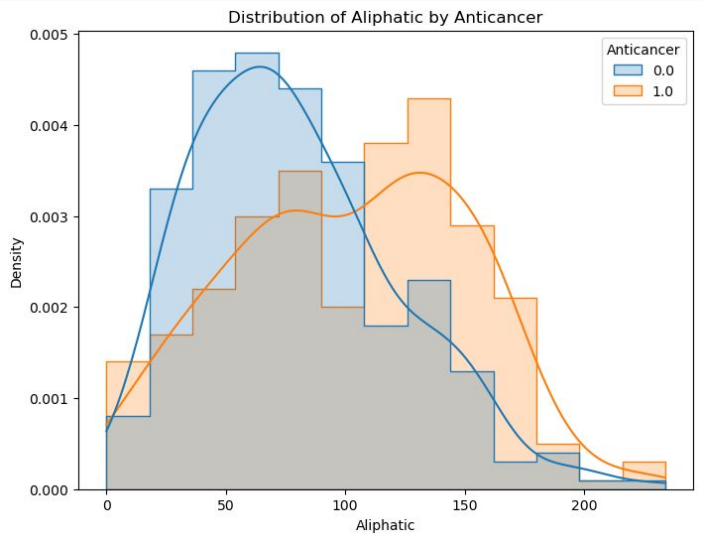




05 Defense

New Peptide #1: VAVERIRILMFELM

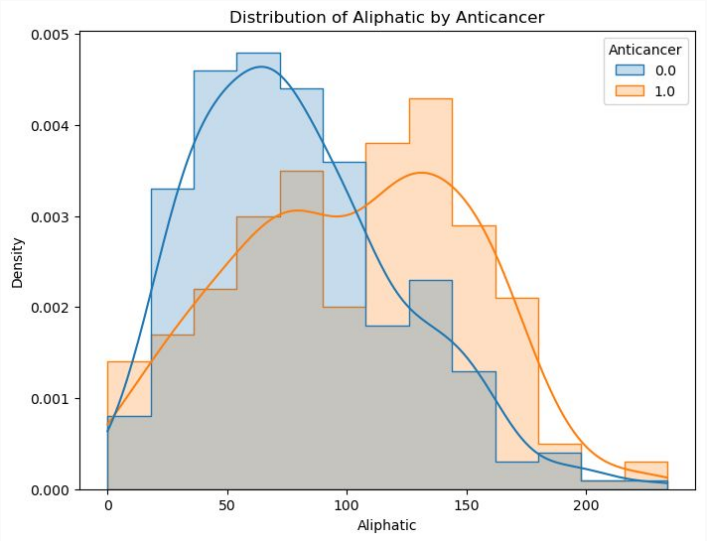
Features



- Charge is positive
- pI ensures that it remains charged in aqueous solutions
- Boman Index of 0.4436
- GRAVY of 1.2429
- Aliphatic Index of 160

New Peptide #2: NNKLMLIWQI

Features



- Charge is positive
- pI ensures that it remains charged in aqueous solutions
- Boman Index of 0.001
- GRAVY of 1.32
- Aliphatic Index of 156

Safety

Toxicity:

We were unable to derive the hemolytic activity from the information in our dataset, so we could not code a model to predict the toxicity.
However, because we know that our peptides in our database are food derived, we have reason to believe that they’re generally safe for human consumption.